The non-clinical package consists of *in vitro* biological activity studies, discussed under quality aspects, and briefly summarised under nonclinical aspects. Four *in vitro* pharmacology studies were also conducted to evaluate similar functional activity of ABP 654 and Stelara relevant to the mechanism of action (MOA) as part of the totality of evidence to support the functional similarity. These *in vitro* pharmacology studies assessed inhibition of IL-23-induced signal transducer and activator of transcription (STAT)3 signalling, IL-12-induced STAT4 signalling, IL-23-induced interferon gamma (IFN-γ) release, and IL-12-induced IFN-γ release in peripheral blood mononuclear cells (PBMCs) from healthy human donors.

No comparative *in vivo* -pharmacology, secondary pharmacodynamic (PD), safety pharmacology or PD drug interaction studies nor *in vivo* pharmacokinetic (PK), toxicokinetic (TK) or toxicology studies were conducted.

## 2.5.2. Pharmacology

### *2.5.2.1. Primary pharmacodynamic studies*

ABP 654 has the same amino acid sequence, formulations, dosage forms, and product strengths with Stelara. The most relevant difference between ABP 654 and Stelara is that APB 654 is manufactured in CHO cell line whereas Stelara is manufactured in murine myeloma (Sp2/0) cell line, and this change has led to some differences in glycans between ABP 654 and ustekinumab (EU). ABP 654 does not contain detectable levels of non-human glycans, such as N-glycolylneuraminic acid (NGNA) and alpha-galactosylation.

Further differences were observed in the levels of high mannose. Compared to the reference medicinal product, ABP 654 shows an increase in high mannose. No difference in the PK profile was observed in clinical study 20190230.

Assays address inhibition of IL-23 and IL-12 mediated signalling, IL-23 and IL-12 receptor-ligand binding, lack of binding to receptor-bound IL-23 and IL-12 on a cell surface, binding kinetics and affinity for IL-23 and IL-12, binding specificity to cytokines related to IL-23 and IL-12, lack of antibody dependent cell mediated cytotoxicity (ADCC), complement dependent cytotoxicity (CDC), and antibody dependent cell mediated cytotoxicity (ADCP) activity, binding to neonatal Fc receptor (FcRn) and first sub-component of complement (C1q), and binding to Fc gamma receptors (FcγRIa, IIa, IIb, IIIa, and IIIb).

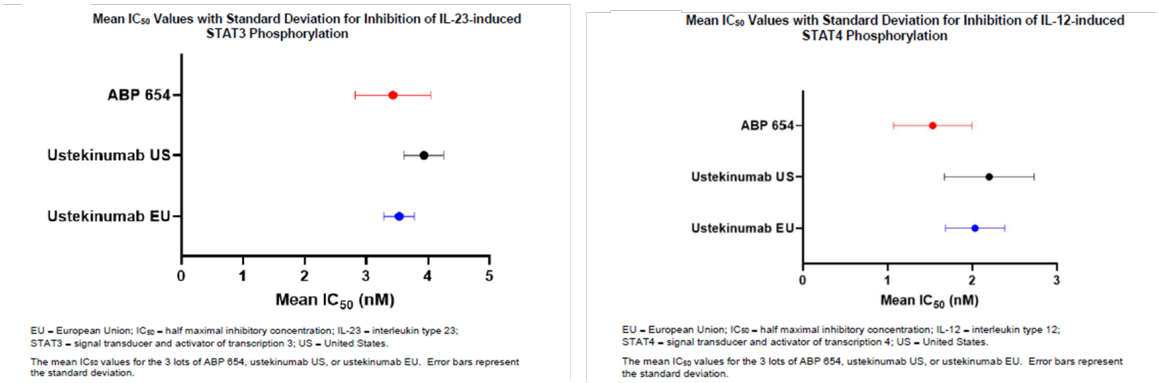
The *in vitro* pharmacology studies to investigate the functional activity of ABP 654 compared to Stelara EU/US assessed inhibition of IL-23-induced STAT-3 signalling, IL-12-induced STAT-4 signalling, IL-23 induced IFN-γ release and IL-12-induced IFN-γ release in activated T-cells from healthy human donors Human peripheral blood mononuclear cells isolated from one donor were used in the experiments.

The formulation of the ABP 654 that was used during the pharmacology programme is representative to ABP 654 clinical SC formulation and identical with RMP Stelara SC formulation. The IV formulation of ABP 654 has a slightly different formulation and is also identical with RMP Stelara IV formulation however has not been included to the pharmacology studies.

The inhibition of the STAT-3 and STAT-4 phosphorylation in activated T cells were investigated by using flow cytometry in three different batches of ABP 654, Stelara EU and Stelara US. The concentration range for tested materials was 100nM to 5pM in 3-fold dilutions. ABP 654 showed slightly lower $IC_{50}$-values of IL-23 inhibited STAT-3 (2.9 - 4.1) in comparison to Stelara EU/US (3.3 - 3.8) with higher standard deviation (0.6 vs 0.3) and lower $IC_{50}$-values of IL-12 inhibited STAT-4 phosphorylation (1.0 - 1.8) in comparison to Stelara EU/US (1.7 - 2.4) with slightly higher standard deviation (0.5 vs 0.4) (Figure 5). The supportive data from an IL-12-induced IFN-γ production assay

demonstrated that ABP 654 is similar to ustekinumab (EU) for inhibition of IL-12-induced IFN-γ production. ABP 654 is within the ustekinumab (EU) quality range, and also in the min-max range – except for one lot.

**Figure 1.** Mean $IC_{50}$ values with standard deviation for inhibition of IL-23 induced STAT-3 and IL-12 induced STAT-4 phosphorylation.




The inhibition of IL-23 and IL-12 induced release of IFN-γ was assessed for ABP 654, Stelara EU and Stelara US in 3 pairwise comparison experiments using one batch of each product. The standard deviation presented is associated with the 3 replicate experiments and does not represent the batch variability but the test conditions. The concentration range for tested materials was 30 - 0.00384 µg/ml (155322) and 90 - 0.0012 µg/ml (155353) in 5-fold dilutions. The homogenous time resolved fluorescence (HTRF) kit was used to measure IFN-γ levels (human IFN-γ kit).

The % relative potency was similar for ABP 654, Stelara (EU) and Stelara (US). The summaries of pairwise comparisons are represented in the Table 8 and Table 9.

**Table 2.** Summary of pairwise comparison on ABP 654, Stelara (EU) and Stelara (US) in inhibition of IL-23 induced IFN-γ release assay.

| | | Relative Potency (%) | | | | | |
|---|---|---|---|---|---|---|---|
| Test Article | | Similarity Assay 1 | Similarity Assay 2 | Similarity Assay 3 | Average | SD | %CV |
| ABP versus EU | ABP 0010485699 | 98.7 | 91.9 | 115.4 | 102.0 | 12 | 12 |
| | EU JES0TML | 95.4 | 110.1 | 94.6 | 100.0 | 9 | 9 |
| ABP versus US | ABP 0010485699 | 84.9 | 101.5 | 82.2 | 89.5 | 10 | 12 |
| | US JCS0WMD | 94.9 | 86.2 | 90.4 | 90.5 | 4 | 5 |
| EU versus US | EU JES0TML | 87.3 | 89.2 | 91.0 | 89.2 | 2 | 2 |
| | US JCS0WMD | 81.1 | 110.4 | 104.8 | 98.8 | 16 | 16 |

CV = coefficient of variation; EU = European/Europe; IFN-γ = interferon gamma; IL-23 = interleukin-23; PBMC = peripheral blood mononuclear cell; US = United States.
Source: ELN ID 20211207-00065

**Table 3.** Summary of pairwise comparison on ABP 654, Stelara (EU) and Stelara (US) in inhibition of IL-12 induced IFN-γ release assay.

| | | Relative Potency to Plate Control | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Test Article | | Similarity Assay 1 | Similarity Assay 2 | Similarity Assay 3 | Similarity Assay 4 | Similarity Assay 5 | Similarity Assay 6 | Average | SD | %CV | Total Valid Runs of Pairwise Comparison |
| ABP versus EU | ABP 0010485699 | 99.1% | 116.4% | — | 87.0% | 98.9% | — | 100.4% | 12% | 12% | 4 |
| | EU JES0TML | 86.4% | 92.9% | — | 87.7% | 91.8% | — | 89.7% | 3% | 3% | |
| ABP versus US | ABP 0010485699 | 84.4% | 88.2% | 111.2% | 87.1% | — | — | 92.7% | 12% | 13% | 4 |
| | US JCS0WMD | 81.3% | 83.5% | 101.5% | 91.2% | — | — | 89.4% | 9% | 10% | |
| EU versus US | EU JES0TML | 110.7% | — | — | — | 105.5% | 111.5% | 109.2% | 3% | 3% | 3 |
| | US JCS0WMD | 107.4% | — | — | — | 97.5% | 118.5% | 107.8% | 11% | 10% | |

— = not applicable; CV = coefficient of variation; EU = European/Europe; IFN-γ = interferon gamma; IL-12 = interleukin-12; PBMC = peripheral blood mononuclear cell; US = United States.
Source: ELN ID 20220118-00020